Skills Summary

Two Forms of the Same Line

Use this reference while completing your worksheet or when studying.

Standard form: $Ax + By = C$ — both variables on the left, a number on the right. Good for budgets and totals, and for reading intercepts.

Slope-intercept form: $y = mx + b$ — solved for y. Good for reading the rate of change and for graphing.

Skill 1 — Standard form to slope-intercept form

Rule: move the x-term to the right, then divide *every* term (both of them) by the coefficient of y. Dividing only one term is the usual mistake.

Example: Write $2x + 5y = 20$ in slope-intercept form.
Step 1. The target form is solved for y, so isolate y — subtract the x-term first, and divide only at the end.
Step 2. $5y = 20 - 2x$, then divide both terms by 5: $y = 4 - \frac{2}{5}x$.
Written in $mx + b$ order: $\boldsymbol{y = -\frac{2}{5}x + 4}$

Try it: Write $3x + 4y = 24$ in slope-intercept form.

check: $\boldsymbol{y = -\frac{3}{4}x + 6}$

(!) Watch out: dividing by the y-coefficient must hit the constant too. From $5y = 20 - 2x$, both the 20 and the $2x$ are divided by 5.

Skill 2 — Slope-intercept form to standard form

Rule: substitute the $mx + b$ expression into $Ax + By$ (choosing A and B so the fraction clears). The x-terms cancel and what is left is the constant C.

Example: Write $y = -\frac{4}{3}x + 6$ as $4x + 3y = C$.
Step 1. The denominator is 3, so multiplying y by 3 clears the fraction — that is why $B = 3$ was chosen.
Step 2. $4x + 3\left(-\frac{4}{3}x + 6\right) = 4x - 4x + 18$.
The x-terms cancel, leaving the constant. $\mathbf{4}\boldsymbol{x} + \mathbf{3}\boldsymbol{y} = \mathbf{18}$

Try it: Write $y = \frac{3}{2}x - 5$ as $3x - 2y = C$.

check: $\mathbf{3}\boldsymbol{x} - \mathbf{2}\boldsymbol{y} = \mathbf{10}$

Skill 3 — Intercepts straight from standard form

Rule: the x-intercept is where $y = 0$, the y-intercept where $x = 0$. In $Ax + By = C$ each substitution kills one term and leaves a one-step equation.

Example: Find both intercepts of $3x + 5y = 15$.
Step 1. Intercepts are the points where the line meets an axis, so set the other coordinate to zero and solve what is left.
Step 2. With $y = 0$: $3x = 15$. With $x = 0$: $5y = 15$.
x-intercept $\boldsymbol{x = 5}$, y-intercept $\boldsymbol{y = 3}$

Try it: Find both intercepts of $2x + 7y = 14$.

check: $\boldsymbol{x = 7}$ and $\boldsymbol{y = 2}$